iLLD-TC4-v2.8.0

Release Notes

Product name: TC4xx_SW_iLLD
Release number and version: iLLD-TC4-v2.8.0
Type of release: Alpha [1]
Release method: Github
Processor platform: TC4Dx AA, TC4Dx AB
Date: 2026-11-16
Previous release number and version: iLLD-TC4-v2.7.0

About this document

Scope and purpose

This document details the release contents, all known issues in the release, and the changes from the last release, together with information on tools and support changes.

New issues identified since the last release of this document are detailed first, followed by all issues identified in previous versions of this release.

Note: [1] This release is not intended for production use. This is reference software that helps the customers to develop their software.

Attention: Refer to the Limitations and deviations chapter before using the software for integration.

Intended audience

This document is intended for anyone using the TC4xx iLLD software.

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1 Release contents

1.1 Release overview

This is an Alpha Release of iLLD SW for the AURIX™ TC4xx platform.

1.2 Released items

This release contains the release items mentioned in Table 1

Table 1 Release content

Folder/File	Description
doc	1. README.md containing link to iLLD documentation 2. Release notes
examples	iLLD template projects
src	iLLD source for TC4Dx device
IFASLL202501.pdf	TC4xx Click License
README.md	Contains the information about the folder structure, package usage and iLLD software release version
scripts	Scripts for generating doxygen documentation
.github	Github workflows file

The Hardware functional blocks supported in TC4Dx are mentioned in Table 2

Table 2

Hardware functional block	iLLD driver
CPU	Cpu, Stm
NVM	Flash
SDMA	Dma
PPUC	Ppuc
GETH	Geth
HSPHY	Hsphy
HSSL, HSCT	Hssl
XSPI	Xspi
PCIE	Ep, Rc
DRE	Dre
MCMCAN	Can, Cre
ERAY	Eray
CAN-XL	Canxl
ASCLIN	Asc, Spi, Lin
SENT	Sent

PSI5	Psi5
PSI5s	Psi5s
MSC	Msc
SDMMC	Sd, Emmc
QSPI	QspiMaster, QspiSlave
I2C	I2c
CLOCK	Clock
VMT	Vmt
SCU	Eru
PMS	Pms
SMM	SmmReset, SmmSysMode
IR	Src
WTU	Wtu
SMU	Smu, SmuStdby
FCE	Fce
eGTM	Tom, Atom, Tim, Trigger, Pwm, Input
ADC	Tmadc, Cdsp, Dsadc, Fcc
LETH	Leth
PORTs	Port
AP	Apu, Prot
SCR	AdcComp, Cpu, Port, Rtc, Scu, Ssc, T2ccu, Timers, Uart, Wcan,I2C
LBIST	Tri

1.3 Compatibility

The iLLD software now includes support for C++ compilation. To accommodate this new feature, several modifications have been made to the make options. For users who are utilizing iLLD in C projects and do not wish to alter their make options, it is necessary to incorporate specific C++ dummy functions into the “Ifx_Cfg_Ssw.c” file to enable successful compilation.

```
#ifdef __TASKING__
void _main(void)
{}
void _doexit(void)
{}
#elif defined(__HIGHTEC__) && !defined(__clang__)
void _init(void)
{}
#elif defined(__GNUC__) && !defined(__HIGHTEC__)
void _init(void)
{}
#elif defined(__ghs__)
void _main(void)
{}
void exit(int a)
{}
#endif
```

Add “atexit” macro in Ifx_Cfg_Ssw.h

```
#define atexit Ifx_atexit
```

Add below lines of code in Ifx_Cfg_Ssw.c file:

```
#if defined(HIGHTEC) && defined(clang)
void __libc_init_array(void)
{}

void __libc_fini_array (void)
{}

int Ifx_atexit(void (*func)(void))
{
return 0;
}

#endif
```

2 Tool information

The tool information is provided in the following table.

Table 3 Tool details

Tool description	Version details
Compiler	Hightec LLVM v11.1.0 TC CTC SmartCode v10.4r1 GreenHills v2026.1 Windriver v5.9.9.1 IFX GCC 2026.03 SNPS Metaware v2.1 TC CARC SmartCode v10.4r1 IFX LLVM (PPU) 21.1.8 Hightec PPU/CDSP - v1.0.3
Processor platform	TC4Dx
Evaluation hardware	TC4Dx_VDK_v2.5.0 TC4Dx: Target hardware (Triboard) TC4Dx: Target hardware (Emulation)

3 Summary of changes

3.1 Issues fixed in release v2.8.0

Issue type	Summary	Description
Bug	Wrong implementation of IfxHsphy_Phy_readMDIORegister() and IfxHsphy_Phy_writeMDIORegister()	[9846] Removed the wrong left-shift assignment to PHYREGi_CTRL.ADDR, which now is assigned by the PHY register address directly.
Bug	[TC4Dx] EGTM TOUT pins are missing in the pinmap file	[9813] The missing TOUT pins “TOUT36, TOUT37” are added for EGTM module.
Bug	TC4xx iLLD CAN reception fails with internal timestamping	[9854] IfxCan_Node_setInternalTimeStamp API disables the configuration change during Node initialization, which is currently re-enabled.
Bug	P12_ACCGRP Missing for TC46 and TC4D	[9857] Missing P12_ACCGRP register instances are added in SFR.
Bug	Spelling issues that codespell found on the illds	[9858] Spelling mistakes have been fixed in the iLLD drivers.
Bug	HighTec LLVM 11.0.0 drops trap tables and BMHD (linker script KEEP rules)	[9877] Updated the linker scripts to include wildcards, ensuring the input sections are no longer ignored.
Bug	Incorrect descriptor element address calculation in IfxPpucStu_initChannel	[9764] Fixed the Incorrect descriptor element address calculation in the API IfxPpucStu_initChannel
Bug	IfxClock_getLeth100Frequency returns wrong frequency	[9743] Fixed IfxClock_getLeth100Frequency() to return the correct LETH100 clock frequency.
Bug	[TC4xx] IfxPort usage of APU APIs leads to a trap when IFX_PROT_ENABLED is not set	[9804] Implemented measures to avoid bus errors on PROT-protected PORT SFR accesses when IfxPort_initApu() is called before other PORT APIs, preventing the PROT state from entering CONFIG mode.
Bug	iLLD Bug in Overlay register configuration (OMASK, and RABR.OMEM)	[9869] OMASK and OMEM related enums are updated as per UM.
Bug	[TC4x Leth]: IfxLeth_setMdioOnlyPins misbehavior	[9579] Enhancement in API and Leth pinmaps for MDC pin configuration for pins controlled by IP. Added union support for Port controlled vs IP controlled pins (H/w Controlled). For MDC pin muxed with RXD pins, support added for RXD select.
Enhancement	Support Hightec LLVM Compiler for PPU	[9654] Added Hightec LLVM compiler support for PPU.
Enhancement	TC4xx add API to get the safety indicator (SAFEIND) and Key-off indicator (KEYIND)	[9809] Two new APIs are added in TRI module to get safety and keyoff indicator.

Enhancement	Introduce new CDSP low level driver	[9784] Introduced new APIs to handle CDSP core operations.
Enhancement	iLLD License text update for open market devices (boost license)	[9859] Replaced the IFASLL license with boost license for TC4Dx for public release.
Enhancement	Make PFwait states dependent on setting of IFX_CFG_CLOCK_SYSPLL_FREQUENCY	[9783] PNVM and DNVM wait states are now calculated based on IFX_CFG_CLOCK_SYSPLL_FREQUENCY.
Enhancement	Implementation of IfxVmt_getSystemAddress for RAM Address Translation in TC4x iLLD	[9779] Added new API getSystemAddress in VMT driver.
Enhancement	TC4xx: Addition of MONBIST in Ssw	[9803] MONBIST APIs are added for SSW.
Enhancement	Implement support for HS Mode for QSPI	[9810] Added support for enablement and configuration of High Speed Mode by the QSPI master.
Enhancement	IfxCan_Can_initNode() checks rxf1f.typeOfService when enabling the rxf1n service request	[9886] During CAN node initialization, condition for configuring the RX FIFO 1 new-message (rxf1n) service request checks rxf1f.typeOfService for DMA instead of rxf1n.typeOfService.
Enhancement	TC4x Pinmaps: Missing pins in multiple modules & devices, regenerate all pinmaps for all devices as per latest Pin Assignement sheet	[9850] TC4x pinmaps are updated to fix the issue of missing pins in multiple devices.
Enhancement	iLLD I2C API: add timeouts	[9918] Configurable timeouts added to while loops waiting for hardware status in read and write APIs.
Enhancement	Test iLLDs for TC4xx for C++ and have a simple demo project included	[9656] CPP Compilation fix is added for ADS project handling
Enhancement	TC4xx: Access protection for Basic projects	[9688] Access protection is added for all template projects
Enhancement	GCC compiler linkers support for CDSP	[9921] Provided Gcc compiler linker support for Cdsp.

3.2 Issues fixed in release v2.7.0

Issue type	Summary	Description
Bug	TC4xx SSW function Ifx_Ssw_Pms_Init()	[9536] Implemented Ifx_Ssw_Pms_Init() and Ifx_Ssw_Pms_InitCheck() in Ifx_Cfg_Ssw.c across all TC4xx BaseFramework and HypervisorDemo projects. The TODO placeholders were replaced with actual calls to IfxPmsEvr_runInitSequence() and IfxPmsEvr_areInitValuesRight() from the existing PMS EVR driver
Bug	ASCLIN: Enable Parity Error Flag is set even when not configured	[9550]Default enablement of parity error notification is removed

Bug	Ifx_SizeT count parameter is wrongly of signed type in IfxAsclin_Asc_write function	[9577] Ifx_SizeT replaced with uint16 and sint32 replaced with uint32 to remove usage of signed data type as return type.
Bug	Issue with QSPI DMA configuration	[9637] In IfxQspi_SpiMaster_initModule(), check against invalid Dma channel index is updated
Bug	TC4 iLLD 2.6 setup issue when relocating project	[9638] Note has been added in RN in the section 6.2 for user to explain the issue and the resolution if the project is relocated to another location
Bug	Missing LPE Interrupt Enable in IfxAsclin_Lin_sendFrame causes driver hang	[9639] Enabled LPE interrupt in IfxAsclin_Lin_sendFrame() function
Bug	Wrong enumeration values with the enum IfxAdc_Select_BoundaryFlag	[9652] Updated enum for IfxAdc_SelectBoundary flag
Bug	CAN Timestamp Pointer Bug	[9690] CAN time stamp validity check added before reading the timestamp from correct timestamp index.
Bug	ASCLIN FIFO initialization ends in Instruction Trap if buffer memory is not aligned	[9717] Added documentation highlighting the need for buffer addresses configured for Tx and Rx buffers to be word-aligned
Bug	Bug in baudrate calculation : IfxMsc_downstreamFractionalBaudCalculator func	[9720] Fixed integer overflow in fractional baud rate calculator
Enhancement	ADC TC4xx : Remove the dependency from Service/Bsp.h file for ADC module and reuse the definition from newly added _Lib/Timer	[9153] Removed the dependency on the Service-layer by using the iLLD-local timer library instead
Enhancement	iLLD - [TC4x][GLS][MAX][SDMMC] Timing issue in High speed mode (50MHz)	[9435] Added additional parameter to configure CMD pin speedGradeCmd in Sdmmc
Enhancement	[iLLD TC4x XSPI] Dangling pointer in IfxXspi_Spi_initModuleConfig leads to unintended dmaConfig value	[9482] Pointer address was wrongly passed in the initModule API of the XSPI
Enhancement	New API/Enhancement request for eGTM TIM	[9530] Added new APIs for EGTM
Enhancement	Free CLANG/LLVM compiler PPU support	[9540] Added the IFX LLVM PPU compiler support for TC4x devices
Enhancement	TC4x CAN CRE: Additional advanced config support in IfxCan_Can_setExtendedUnicastRouting	[9551] Missing CAN Frame format modifiers added to support unicast routing of extended can frame Ids
Enhancement	improvement request for clock configuration	[9685] Added a configurable osConModeVal field to IfxClock_Config so the oscillator mode (OSCCON.MODE) can be selected via the configuration structure instead of relying solely on the IFX_CFG_CLOCK_EXT_CLOCK compile switch. IfxClock_powerOnCrystalOsc() now takes

		the mode as a parameter and programs OSCCON.MODE from osConModeVal during PLL/clock init.
Enhancement	IFX LLVM compiler support for PPU for all TC4xx device	[9698] Added IFX LLVM compiler support for PPU
Enhancement	Compiler warning fix for Alpha release	[9707]No new line at the end of the file warning is fixed in all the base template projects

3.3 Issues fixed in release v2.6.0

Issue type	Summary	Description
Bug	Wrong header file name	[9397] The inclusion path is updated to "<sfr/regppu.sfr>" resolving the issue for PPU build when using TASKING compiler.
Bug	STM iLLD driver update required towards STM trap issue along with compilation issues	[9322] Updated Stm driver (IfxStm_initCompare) to use IfxStm_getIndex API instead of CPUx_HRA_CORE_ID
Bug	Mpu Configurations: Hypervisor software needs correction	[9256] Corrected data ranges to point to data memory sections
Bug	[TC4xxx][TMADC] Issue's with group functionality and readings from result registers	[8763] Fixed bus trap issue and documented a limitation of 7 channels per group when timestamp is enabled due to AW2 memory constraints.
Bug	Internal remarks exist in TC4xx iLLD source code	[9436] Code cleanup by removing internal remarks in the iLLDs
Bug	Issue with (e)GTM bugs	[9480] Fixed ISR priority check in interrupt configuration. Previously, setup only proceeded when isrPriority equaled 1, blocking other valid priorities. The condition now uses if (config->isrPriority), allowing any non-zero priority value.
Enhancement	VMT nondestructivetest api update	[9352] Updated the runNonDestructiveInversionTest API by writing 0x0009 to CONTROL register to start the test in the api
Enhancement	TC4x CDSP: Input address automatically updated by the init function	[8496] CDSP driver code has been updated to support configuration of input source address for FFT filter type
Enhancement	API request in eGTM	[9203] Api's IfxGtm_Tbu_disableChannel() and IfxGtm_Tbu_setChannelBase() are Added for eGTM and Gtm
Enhancement	Provide an API to update frequency in (E)GTM (A)TOM_PWM driver	[9238] Added API to update frequency in (E)GTM (A)TOM_PWM driver
Enhancement	Provide an API to update dutycycle in (E)GTM (A)TOM_PWM driver	[9236] Added API to update duty cycle in (E)GTM (A)TOM_PWM driver

Enhancement	TC4xx CDSP initDspCore function is loading DCCM image twice in case of Lo filter chain (FC5)	[9040] Updated IfxAdc_Cdsp_FilterChainSel enum for custom image. Updated iLLD documentation.
Enhancement	provide an empty implementation of allowAccess() and make it a weak function that can be overridden from user application	[9400] Added weak empty implementation of allowAccess()
Enhancement	API to get the signal level of input to a TIM channel filter	[9329] Added API to get the signal level of input to a TIM channel filter
Enhancement	API to enable Bit Compression input to the Tim_In channel	[9326] Added API to enable Bit Compression input to the Tim_In channel
Enhancement	Provide API to update GPR1 register in Gated Periodic Sampling mode	[9334] Added API to update GPR1 register in Gated Periodic Sampling mode
Enhancement	Two APIs to (temporarily) suspend and restore all clocks of (e)GTM.CMU	[9333] Added two APIs to suspend and restore all clocks of (E)GTM.CMU
Enhancement	PWM Driver: Provide an API to enable self reset of fast shutoff status	[9331] Added API to enable self reset of fast-shutoff and an API to reset fast-shutoff for (E)GTM PWM
Enhancement	Provide an API to get TIM input signal status.	[9330] Added API to get TIM input signal status.
Enhancement	API to get bit compression data from Tim_In channel	[9327] Added API to get bit compression data from Tim_In channel
Enhancement	Provide enable and disable APIs for Tim_In channels	[9335] Added APIs to enable and disable the Tim_In channels
Enhancement	Provide an API to connect boundary flag of (TM/EV)ADC sub units to (e)GTM	[9336] Added APIs to connect boundary flag of (TM/EV)ADC sub units to (e)GTM
Enhancement	New optimized API to connect (e)GTM.TIM to ADC Boundary Flag signals`	[9325] Added New optimized API to connect (e)GTM.TIM to ADC Boundary Flag signals
Enhancement	Provide an API to reset fast Shutoff by software	[9332] Added API to enable self reset of fast-shutoff and an API to reset fast-shutoff for (E)GTM PWM
Enhancement	Provide API to get TIM filter output signal status.	[9328] Added APIs to get TIM filter output signal status
Enhancement	IfxGeth_Eth_shuffleRxDescriptor: improved version of tail pointer	[8043] Changes done in multiple APIs to support extended use cases as per AE feedback. Does not affect existing applications functionally. 1. IfxGeth_Eth_initReceiveDescriptors: Reduction in number of Rx Descriptors initialised by 1. Additionally, explicit initialisation of rx tail descriptor RDES3 to 0. 2. IfxGeth_Eth_initTransmitDescriptors: Addition of initialisation of TXDESC_TAIL_LPOINTER register to explicitly synchronise H/w also during init.

		3. IfxGeth_Eth_shuffleRxDescriptor: API updated to release tail descriptor instead of current descriptor.
Enhancement	License update (click & file header)	[9404] The boost license text is updated with the latest IFASLL license.
Enhancement	TC4x PPU: Support Mutex and Semaphore	[9421] Added mutex and semaphore support for PPU

3.4 Issues fixed in release V2.5.1

Issue type	Summary	Description
Bug	QSPI communication issue : 2.5.0	[9467] Reverted the previous changes to the logic on setting of Clock Phase to the ECON register. User document updated with enums used for clock phase configuration and the resultant register bit-fields written. Qspi slave initialization defaults to CPH value of 1

3.5 Issues fixed in release V2.5.0

Issue type	Summary	Description
Bug	TC4Dx PPU Clock config wrong	[8896] Added support for K3PREDIV and K3DIV support in SYSPLLCON1 register to derive the ppupll clock.
Bug	Possible bug in IfxVmt.c TC4xx iLLD 2.4.0-EIR	[9070] Removed unwanted masking which writes to reserved bits when accessing MCONTROL register in IfxVmt_readSramAddress and IfxVmt_writeSramAddress API
Bug	[TC4xx][CDSP]: Using GCC compiler with O3 optimizations enabled leads to a trap when clearing CDSP memories	[9112] Added Volatile keyword to prevent the optimization
Bug	EGTM TIM confusing enum elements	[9241] Enums are renamed to better reflect its function from a signal point of view
Bug	I2C: No handling for FIFO Underflow Errors in I2c_read and I2c_write APIs	[9255] FIFO Underflow handling support added for read and write APIs
Bug	Instruction trap caused by HRA_CORE_ID reading without mfcr	[9271] Updated IfxStm_initCompare Api to get index from passed cpu parameter using different way rather than using HRA_CORE_ID which used to cause trap
Bug	TC4x HSPHY XPCS: IfxHsphy_Xpcs_setSgmiiSpeedMode seqeunce different compared to UM	[9279] Initialization sequence updated for XPCS
Bug	xSPI: Fix IfxXspi_Status enum	[9346] Removed IfxXspi_Status_noError as its redundant

Bug	IfxCpu_Trap_extractTrapInfo not returning the expected values in each core while using GCC	[8519]IfxCpu_Trap_extractTrapInfo not returning the expected values in each core while using GCC compiler
Bug	[TC4xx][CDSP]: Wrong usage of CDSP result register RES0 in IfxAdc_Cdsp_readDspCoreResult	[8753] Returned the valid result from register if event flag is not set
Bug	IfxCanxl_Can_stopNode is not using the correct LOCK register	[8899] Register MH_LOCK used for unlock sequence over Ni_LOCK
Bug	xSPI: Wrong ARLEN and AWLEN configuration in IfxXspi_Spi_dmaEnhancedDataExchange	[8906] The configuration of ARLEN and AWLEN burst lengths for internal DMA is updated to use correct TMOD
Bug	CPU Generic File includes Tricore specific file	[8993]XSPI driver is moved to cpugeneric from tricore, due to dependency for hsphy which is present in cpugeneric
Bug	bug for get the frequencies	[9007] Fixed the rounding off issue in IfxClock_getPllFrequency, IfxClock_getPerPllFrequency1, IfxClock_getPerPllFrequency2, IfxClock_getPerPllFrequency3 APIs
Bug	Linker script of TASKING for PPU needs correction.	[9056] The linker script is updated for PPU for defining the stack size
Bug	CompilerHightec.c creates compilation issue in PPU	[9064] Conditional check if added to include the ssw file only when compiling for tricore and avoid when building ppu
Bug	ASClIN refers to Services files	[9065] Dependency with service layer removed for ASCLIN
Bug	Not all system services are compilable for PPU	[9066] The SysSe folder is located in the CPUGeneric folder but COMM and MATH libraries inside it are not compilable for PPU either they need to be moved out of CPUGeneric or made compilable for all ISAs
Bug	IfxCan_Can_readTxEvent incorrect timestamp reading	[9068] Timestamp read logic updated to use E1B register
Bug	IfxCan_Can_initNode() does not initailize timeStampType of IfxCan_Can_Node	[9069] Configuration for timestampType is updated in module handle for correct reference during initialization
Bug	Ambiguity and Potential Misuse of IfxCan_FilterType_none in Extended Filter Configuration	[9257] Documentation updated with correct details about Basic and Extended filter types
Enhancement	EGTM PWM driver interrupts cannot be configured externally	[4549] Updated PWM interrupt configuration to check for priority instead of callback pointers. This change provides more flexibility and ensures consistency across the Egtm modules.

Enhancement	TC4x LETH: General update for Optimizing APIs in IfxLeth_Eth.c	[8328] Ethernet driver requires optimal implementation due to high-speed nature by reduced number of access (read/write) to the volatile register for the below APIs
Enhancement	[Linker files]: Unified Linker file design across different toolchains	[8375] Linker files updated for GCC and PPU Metaware in BaseframeworkTC4Dx
Enhancement	Request for Enhanced CRC32 Interface Implementation in ILLD for TC489	[8377] Enhanced CRC32 Interface Implementation in ILLD to : * Support data length up to 32-bit range * Configurable endianness handling via compile-time switch * Custom seed value initialization * Multi-compiler compatibility
Enhancement	solve access data no alignment by using memcpy api to copy data to destinationAddress IfxCan_Node_writeTxBufData,IfxCan_Node_writeData	[8521] 32-bit data copy replaced with bytewise data copy to prevent unaligned address access
Enhancement	TC4x Enhance the CAN loopback mode for LOUT feature	[8662] Support for LOUT feature has been added, similar to Loopback mode
Enhancement	Add a dummy trap handler to the TASKING PPU project.	[8738] Added default handler for tasking ppu
Enhancement	TASKING PPU Linker Scripts improvements	[8739] Added default handler and linker improvement for tasking ppu
Enhancement	Add AllowAccess to all template projects	[8740] Template project configuration updates done for all PPU devices
Enhancement	[TC4xx][HSPHY]: IfxHsphy_Geth_setupRgmiiOutputPins API configures the TX RGMII pins with the wrong speed grade	[8781] Speed grade updated for RGMII output pin configuration
Enhancement	Add IfxPpucCore_configureAccessToPpuVccm	[8789] Added API IfxPpucCore_configureAccessToPpuVccm() and IfxPpucStu_configureAccessToPpucStuDmi()
Enhancement	[TC4xx] I2C: DMA Support Request	[8808] APIs to support FIFO configuration for use with DMA
Enhancement	TC4x: 10 MHz XTAL clock support shall be removed for external customers	[8813] Removed the 10MHz XTAL support for external iLLD
Enhancement	iLLD for the CDSP GCC Compiler	[8853]Added support for CDSP illds for GCC compiler
Enhancement	allowAccess() function shall be called in Ifx_Cfg_Ssw.c::Ifx_Ssw_AP_Init function	[8963] The "allowAccess" function has been added for the TC4xx device and is now being called from the Ifx_Ssw_AP_Init function in Ifx_Cfg_Ssw.c file

Enhancement	Collect the files from the _Impl folder according to the device	[9020] Collected the config files in _Impl folder into subfolders according to the device
Enhancement	[TC4x][CPU] Unable to sync only specific cores in multi-core environment	[9043] The implementation of the IfxCpu_waitEvent doesn't not allow to synchronize only few specific cores (e..g. Core 2 versus Core 3)
Enhancement	[TC4x][STM] iLLD doesn't allow to init STM of another core	[9045] Updated IfxStm_initCompare Api to get index from passed cpu parameter
Enhancement	Consolidate IfxAdc_XXX_YYYSrvReq type which has same elements	[9046] Updated a common type structure IfxAdc_SrvReq in IfxAdc.h and utilized the same in different ADC files
Enhancement	IfxCpu_configureAccessToCpus give access to CPUCS creating issue with AURIX VP	[9047] Conditional compilation added for IfxApApu_init of CPUCS in IfxCpu_configureAccessToCpus
Enhancement	IfxStm_configureAccessToStms give access to CPUCS creating issue with AURIX VP	[9048] Conditional compilation added for IfxApApu_init of CPUCS in IfxStm_configureAccessToStms
Enhancement	TC4x EGTM: TIM Driver updates needed for adding LUT to input source	[9152] Added LUT feature to Egtm Tim
Enhancement	QSPI: Provide APIs to get the QSPI error information	[9161] Added API to get the error status
Enhancement	QSPI: APIs to get and clear transmit/receive interrupt flags	[9163] Separate APIs ptovided to read and clear the interrupt flags
Enhancement	Update API IfxCan_Can_initNode() to initailize timeStampType of IfxCan_Can_Node	[9344] APIs have been updated to handle the timestamp value correctly

3.6 Issues fixed in release V2.4.1

This is the first public release for the product, hence no issues are listed.

4 Known Issues

The known issues are listed in the below table

Table 4

Key	Summary	Impact	Workaround
-	-	-	-

5 Limitations and deviations

This chapter describes the limitations and deviations due to software/hardware design constraints.

- None

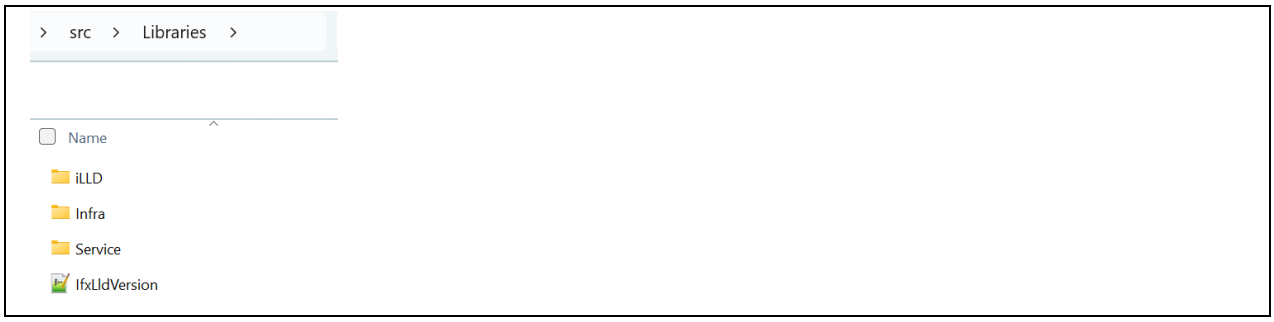
6 Installation

The iLLD software release package is an archive, and it may be extracted to any drive of the computer. There is no specific directory structure needed.

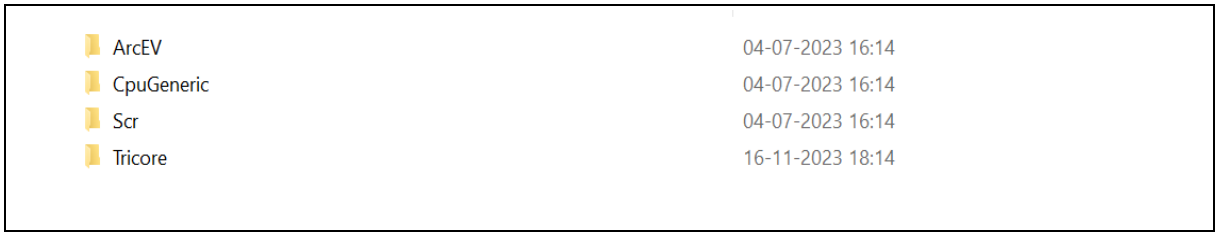
6.1 Package usage

- Package contains the following items.
  - doc : Contains the README.md and Release notes
  - examples : Contains the template project for TC4Dx devices
  - src : Contains Source Code for the TC4Dx drivers
  - IFASLL202501.pdf : TC4xx Click license
  - scripts : Scripts for generating doxygen documentation
  - .github : github workflow files
  - README.md : Contains information about the folder structure, package usage and iLLD software release version

  The src folder contains the source code
  - src/Libraries/
    - iLLD : Directory contains the drivers for peripherals of different architecture
    - Infra : Directory contains sfr's, startup software and platform files
    - Service : Directory contains service layer libraries



  - src/Libraries/iLLD/TC4xx/
    - ArcEv : Directory contains drivers for ArcEv
    - Scr : Directory contains drivers for Scr
    - Tricore : Directory contains drivers for Tricore
    - CpuGeneric : Directory contains drivers common for both Tricore and ArcEv



- The iLLD package contains drivers intended for multiple targets(Scr, Tricore, ArcEv). Template projects listed below are provided as part of package.

Target	Template
Tricore	BaseFramework_TC4Dx BaseFramework _TC4Dx_Scr BaseFramework_TC4Dx_Ppu HypervisorDemo_TC4Dx

Guidelines for SCR Baseframework

Within the SCR package, the i51 core drivers are exclusively compiled using the Tasking compiler tool. The Tricore, on the other hand, is compiled using the supported compiler tool that is currently available.

Once the compilation process is completed without errors, the next step involves flashing the Tricore elf. The Tricore will load a C array into the XRAM memory. This array contains all the necessary information required for the SCR boot-up process.

Guidelines for Ppu Baseframework

The PPU drivers are compiled using the MetaWare Compiler. It is essential to verify the presence of predefined libraries for the TC4xx derivatives. If these libraries are included in the current version of MetaWare, you may proceed with the compilation process.

However, if the predefined libraries for the TC4xx derivatives are not available in the installed version of MetaWare on your system, please follow the steps below to install them:

1. Initiate the command prompt by pressing the combination of the Windows key + R.
2. Input the following command: *cmd /V:ON /k C:<MetaWare-path>setup.bat*. This command will activate the Python environment that is bundled with MetaWare.

To download the predefined libraries of a specific derivative, use the following command:

build -ca -sc < derivative>_system -e BUILD_CONFIG=< derivative>_PPU -e BUILD_TYPE=debug

For instance, for the 4Dx derivative, the command would be:

build -ca -sc tc4Dx_system -e BUILD_CONFIG=TC4D_PPU -e BUILD_TYPE=debug

This command will initiate the download of the predefined libraries for the specified derivative.

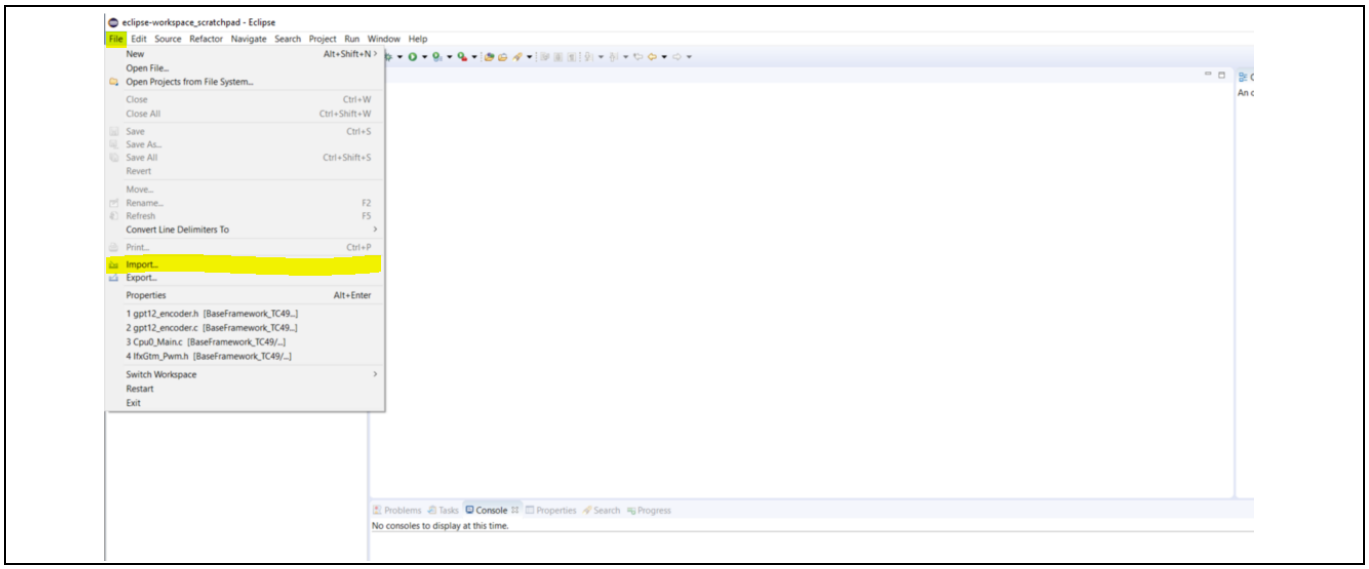
6.2 Importing Baseframework to Bifaces

- Import the template project from the package extracted locally

  Ex: C:\Infineon\iLLD\TC4x\<version>\examples

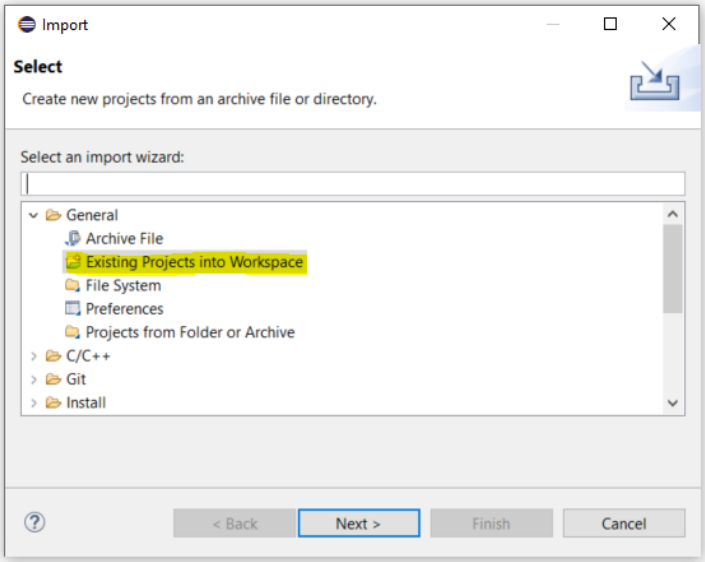
- Open Bifaces. Click on File --> Import

.

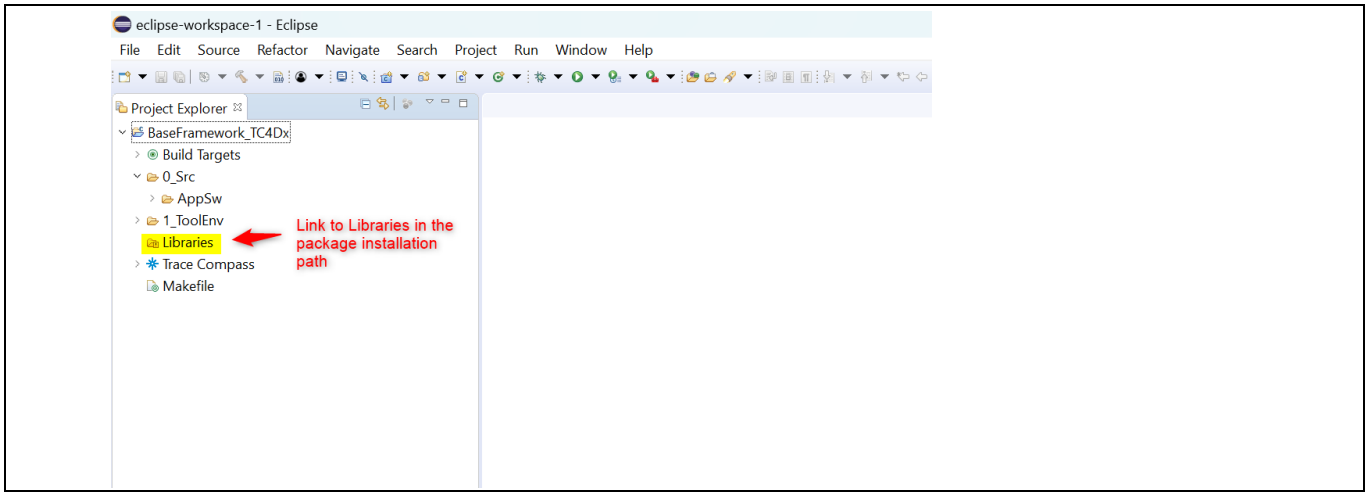


- Select existing project into workspace and Click Next.

.



- Click on Browse and select the root directory where the Baseframework is present and Click ok.



Note:

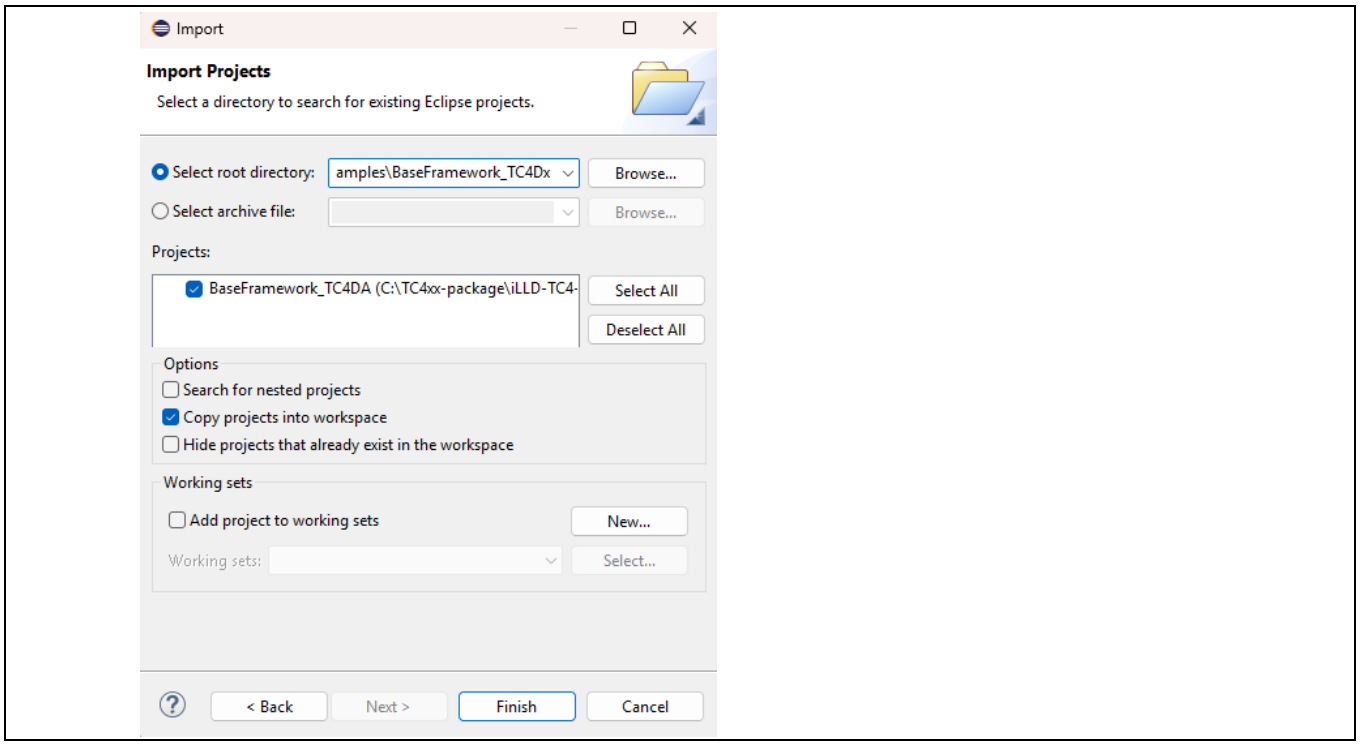
1) The libraries are not included in the template projects, with the exception of the Hypervisor Baseframework. Instead, the libraries are located in the package path and are linked to the template project. This means that after importing the template project into BIFACES, users can compile the project as usual, and the required libraries will be automatically linked..

2) If a template project is copied into a workspace or moved from its default path, the build may fail because the library locations are hardcoded in config.xml, and the project can no longer resolve them. To avoid this, the user must either copy the required libraries into the new project location or update the absolute paths in config.xml accordingly..

- If you want to copy the Baseframework_TC4Dx into workspace, check the box copy project into workspace. Click Finish and the project will appear in Project Browser.

Note: The libraries should also be copied to workspace along with the Baseframework.

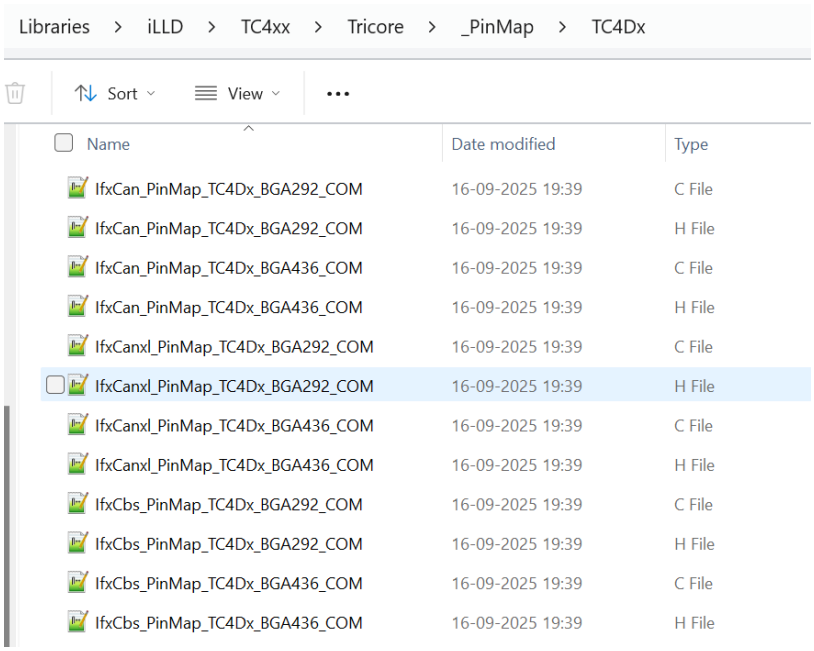
.



Note: The iLLD package contains drivers intended for multiple targets (Tricore, SCR, ArcEv). To compile the drivers, please use the BaseFramework provided as reference, or ensure that drivers intended for SCR target are not built for Tricore.

6.3 Selection of Pin package

- PinMap directory contains all the pinmaps for all the packages I.e. there are no separate folders for different packages.



- To select between the different packages, follow the below steps.
    - Packages can be selected with the help of the macros mentioned below, these should be defined in Ifx_Cfg.h
    - For example, in TC4Dx there are two packages available, so define macros as mentioned below
    - To use the BGA292_COM package, set the value of macro IFX_PIN_PACKAGE_BGA292_COM to 1

      #define IFX_PIN_PACKAGE_BGA292_COM 1
    - To use the BGA436_COM package, set the value of macro IFX_PIN_PACKAGE_BGA436_COM to 1

      #define IFX_PIN_PACKAGE_BGA436_COM 1

Note: Only one macro should be defined in Ifx_cfg.h as explained in below screenshot.

File: Ifx_Cfg.h

```
/* Change Package by defining macro */
#define IFX_PIN_PACKAGE_BGA292_COM 1
// #define IFX_PIN_PACKAGE_BGA436_COM 1
```

6.4 Compiler Selection to build iLLD

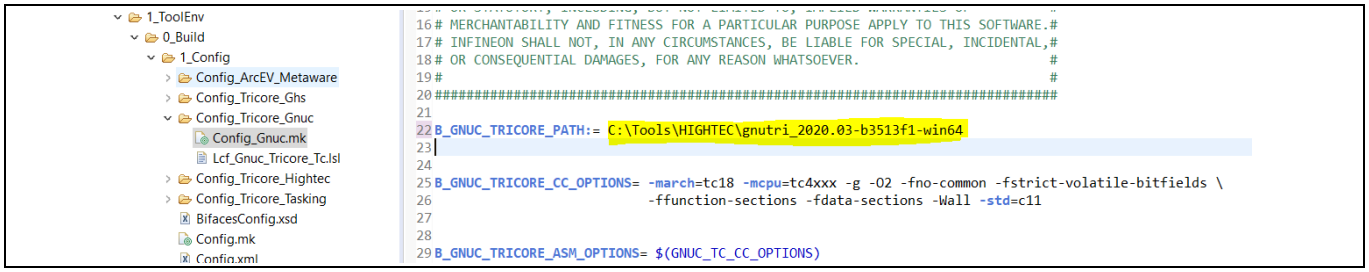
- Multiple compiler toolchains (mentioned in Table Tool information) are supported to build Tricore iLLD's
- Compiler selection is handled in Config.xml present in '**1_ToolEnv\0_Build\1_Config\Config.xml**'

.

```
<!-- Architecture congfigurations -->
<architecture name="Tricore" primaryToolchain= "Gnuc">
    <!-- Target congfigurations -->
    <target name="Tc" enable="true">
        <output type="elf" enable="true" fileName="$(PROJ_NAME)_Tc.elf" tool="Ld"/>
        <output type="hex" enable="true" fileName="$(PROJ_NAME)_Tc.hex" tool="Hex">
            <input siblingTypes= "elf" useObjects="false" />
        </output>
        <output type="srec" enable="false" fileName="$(PROJ_NAME)_Tc.srec" tool="Srec">
            <input siblingTypes= "elf" useObjects="false" />
        </output>
        <output type="lib" enable="false" fileName="$(PROJ_NAME)_Tc.a" tool="Ar"></output>
    </target>
```

6.5 Compiler Path to build iLLD

- Compile path is specified in '1_ToolEnv\0_Build\1_Config\Config_Tricore_{compiler}\Config_{compiler}.mk'



7 Support packages

No Support packages are provided in this release

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